Ahmedabad University

CSE400 Fundamentals of Probability in Computer Science

Section -1

Computer Networks

Group 16

Milestone 4

Optimizing UAV deployment for Multi-UAV Cooperative Sensing

Project Objective and Deterministic Baseline

Baseline Strategy:

- Fixed UAV placement
- Uniform spacing

Working:

- Nearest target selection
- Fixed orientation

Metric:

- Total detection probability

Limitations:

- No adaptability
- Single configuration
- Suboptimal outcomes

System Pipeline Overview

Pipeline:

- Target generation
- UAV deployment
- Signal computation
- Detection probability
- Performance evaluation

Decision Stage:

- UAV placement & orientation

(Key Point: Same pipeline,Different strategy)

Sources of Uncertainty

Noise:

- Random signal variation

Interference:

- Signal overlap

Target Location:

- Spatial randomness
- Targets can be seen in several places.
- Creates confusion in the signals that are received and detected

Impact:

- Uncertain detection
- Variable performance

Key Random Variables

- **Noise:** Gaussian distribution, for fixed: $N \sim \text{Normal}(0, \sigma^2)$
- **Target Signal:** Uniform distribution, for fixed: $S \sim \text{Normal}(0, \sigma s^2)$

Signal Model:

- Signal + Noise

UAV Variables:

- Position, orientation (randomized)

Insight:

- System is probabilistic

Randomized Algorithm Design

Approach:

- Random initialization
- Search-based optimization

Methods:

- PSO → iterative updates
- GA → evolution-based

Randomness:

- Sampling + mutation
- Objective: Maximize detection

Key Difference:

- Baseline: fixed
- GA/PSO: adaptive search

Probabilistic Reasoning Behind Algorithm

- Better UAV placement means higher SNR and higher Pd

Randomization allows:

- Exploration of multiple placements
- Avoidance of poor fixed solutions

Fusion Rule:

$$P_n = 1 - \prod_{m=1}^{M} (1 - P_{m,n})$$

- Cooperative detection improves overall performance

Implementation of Randomized Strategy

Code Components:

- EBD Model : computes SINR and Pd
- Fitness Function : evaluates UAV placement
- PSO Algorithm : optimizes positions

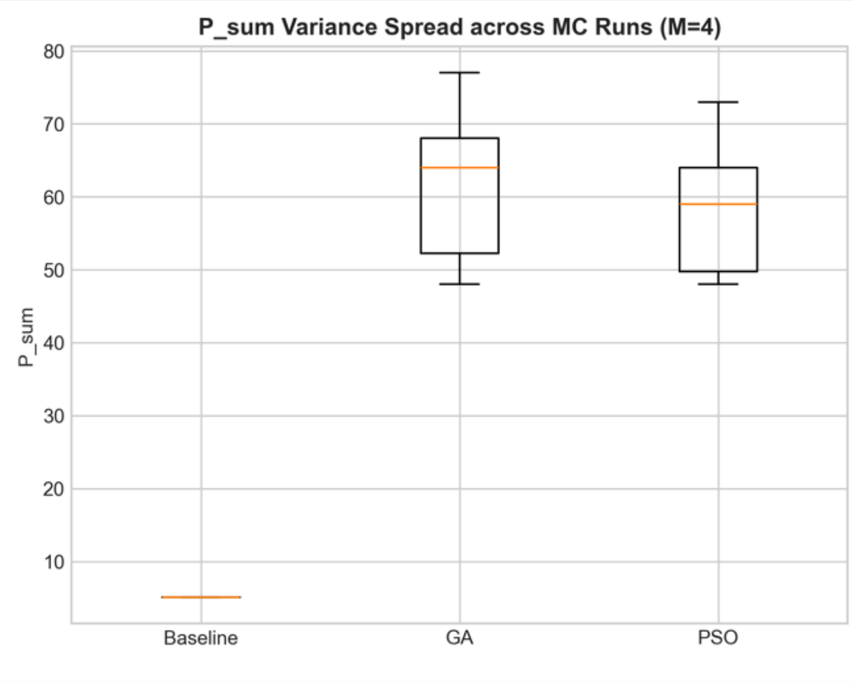
Workflow:

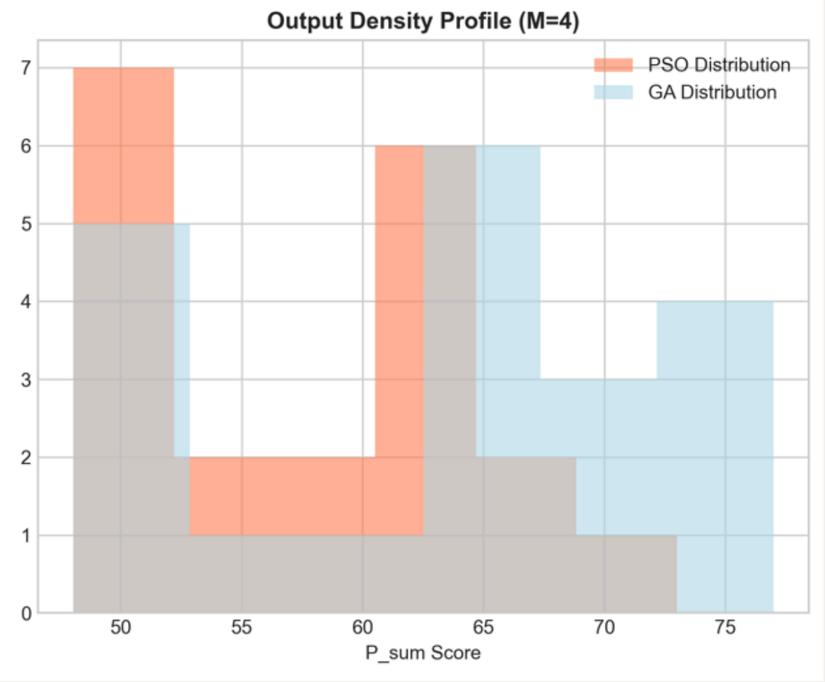
Initialize → **Random placement** → **Evaluate** → **Update** → **Repeat**

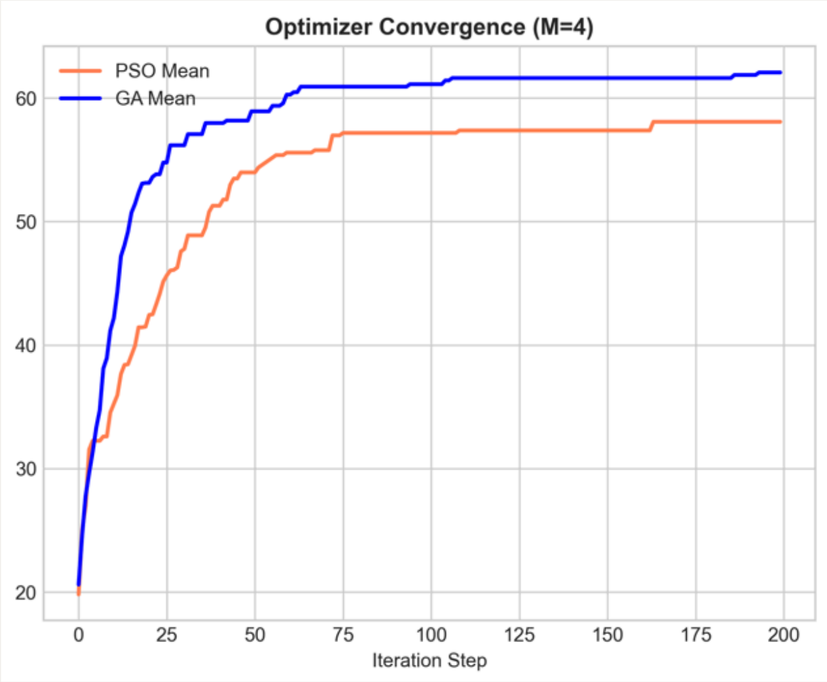
Evaluation Metrics

- Probability of Detection (Pd) maximize
- Fitness Score overall system performance
- Directly measures detection performance
- Reflects system objective
- P_sum (Total Detection Probability) used as main performance metric
- Variance across runs used to measure stability
- Convergence used to evaluate optimization efficiency

Comparative Results







- This plot shows the distribution of total detection performance (P_sum) across multiple simulation runs
- The deterministic baseline gives a fixed value (~5) → no variability
- GA and PSO show higher median performance (~60–65)
- Randomized methods introduce variability but significantly improve performance

- Shows how often different P_sum values occur
- GA distribution is shifted to the right → higher performance values
- PSO slightly lower but still better than baseline
- Demonstrates consistency of randomized algorithms

- Shows how performance improves over iterations
- GA converges faster and reaches higher value (~62+)
- PSO converges slower (~58)
- Both outperform deterministic baseline significantly

Key Insights

- Randomized strategies clearly outperform the deterministic baseline in detection performance
- Genetic Algorithm (GA) achieves better results compared to PSO in our simulation
- Introducing randomness helps in better exploration of the search space
- Optimization algorithms improve performance progressively over iterations

Final Insight:

- Randomized algorithms perform better in uncertain environments

Project Context and Motivation

Project: Optimizing UAV Deployment under Uncertainty

- Domain: UAV sensing & surveillance

Why uncertainty exists in the system:

- Maximize detection probability in randomized algorithm
- Reliable multi-UAV sensing

Why Uncertainty Matters:

- Noise + interference
- Unknown target locations
- Probabilistic detection